

# The problem with localization of gravitational energy in general theory of relativity

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**Abstract:** Using the principle of least action in the general theory of relativity, the equation for metric and the equation for motion of charged fluid particles are derived, as well as the formula for energy of a spherical body with a fluid-like substance. The calculation shows that there is no contribution from the gravitational field in the energy outside the body. Thus, the problem of localization of the gravitational field energy in space arises. In order to solve this problem, it is proposed to replace the scalar acceleration field and the scalar pressure field, which are represented in the Lagrangian density, with corresponding vector fields.

**Keywords:** Lagrangian density; principle of least action; general theory of relativity; equation for metric; equation of motion; energy.

## 1. Introduction

There are many works that analyze the foundations of the general theory of relativity (GTR) and provide corresponding interpretations of the discovered facts. As a theory that successfully describes gravitational phenomena through the geometric properties of space-time, GTR confirms that the space-time continuum, through its static and dynamic fields, actively interacts with the matter particles of physical systems and transfers interactions from one body to another.

Despite the successful predictions of GTR, this theory requires additional work by theorists due to several unresolved issues. For example, it remains unclear how to calculate the contribution of the gravitational field's energy to the relativistic energy of a physical system [1]. In GTR, it is necessary to derive and justify constraints on the theory's parameters, beyond which the predictions of the theory become invalid. This applies, for example, to the appearance of singularities in black holes [2], as well as dark matter and dark energy in the Lambda-CDM model in cosmology [3].

One of the simplest objects considered in GTR is a solitary spherical body consisting of matter in the form of a fluid. Using covariant approach, in [4] various options of the Lagrangian

density were considered, which make it possible to derive from the principle of least action the equations of motion of particles in an ideal fluid in the absence of electric charges of particles and dissipation of energy of the particles. At the same time, it is indicated in [5] that description of a real fluid as an ideal isentropic rotation-less fluid using scalar functions in the Lagrangian density is possible only for a limited period of time.

The purpose of our work is to present the covariant equation of motion for fluid particles inside a spherical body in the general theory of relativity, in the case where the fluid particles have an electric charge. In addition, a formula for determining energy is derived. In Section 2, the equation for the metric is obtained from the principle of least action, followed by the equation of motion for the particles. In Section 3, the equation of motion for the particles is derived directly from the principle of least action, in the form of a balance equation for force densities. Section 4 defines the scalar curvature and cosmological constant in the space outside the body, and Section 5 calculates the energy outside the body. The problem of localizing gravitational energy and solutions to this problem are discussed in Section 6.

## 2. Equation for metric and equation of motion

Let us write in general the action function  $S$  for an arbitrary physical system:

$$S = \frac{1}{c} \int \mathcal{L} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \int \mathcal{L} \sqrt{-g} dt dx^1 dx^2 dx^3. \quad (1)$$

In (1),  $c$  is the speed of light,  $\mathcal{L}$  is the volume density of the Lagrangian function,  $g$  is the determinant of the metric tensor  $g_{\mu\nu}$  with metric  $(+---)$ , and four-dimensional coordinates  $x^\mu = x^0, x^1, x^2, x^3$  are used, where  $dx^0 = c dt$ .

Let there be a continuously distributed matter in the form of a fluid or liquid, with invariant charge density  $\rho_{0q}$  and invariant mass density  $\rho_0$ . The charge density  $\rho_{0q}$  and the mass density  $\rho_0$  are some functions of the state of the matter and may depend on the coordinates and time.

In [6], the elastic energy of a fluid was considered using the principle of least action, and the equation of motion for uncharged fluid particles was derived within the framework of the general theory of relativity. Similarly, for the Lagrangian density  $\mathcal{L}$  of a physical system with the addition of terms responsible for electromagnetic field in a charged fluid, the following expression can be written:

$$\mathcal{L} = ckR - 2ck\Lambda - c^2\rho_0 - \rho_0\Pi - A_\beta j^\beta - \frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta}. \quad (2)$$

Equation (2) contains the expression  $-c^2\rho_0 - \rho_0\Pi$  that, according to [6-7], is necessary to derive equation of motion for the barotropic perfect fluid.

In (2)  $R$  is the scalar curvature;  $\Lambda$  is the cosmological constant;  $\Pi$  is the potential energy of elastic compression of a liquid per unit mass;  $A_\beta$  is the electromagnetic four-potential; the charge four-current  $j^\beta = \rho_{0q} u^\beta$  depends on the charge density  $\rho_{0q}$  and on the particle four-velocity  $u^\beta$ ;  $\mu_0$  is the magnetic constant;  $F_{\alpha\beta}$  is the electromagnetic field tensor.

The constant  $k$  in (2) is determined by the gravitational constant  $G$  and the constant  $\beta$ , which is close to one in absolute value:

$$k = -\frac{c^3}{16\pi G\beta}. \quad (3)$$

The specific energy  $\Pi$  of elastic compression in (2) is determined by the formulas:

$$\Pi = \int_0^P \frac{dP}{\rho_0} - \frac{P}{\rho_0}, \quad d\Pi = \frac{dP}{\rho_0} - d\left(\frac{P}{\rho_0}\right) = \frac{P}{\rho_0^2} d\rho_0. \quad (4)$$

In (4),  $P$  is the scalar pressure, and the mass density can depend on the pressure as a function  $\rho_0 = \rho_0(P)$ . For the mass density differential  $d\rho_0$ , we have  $d\rho_0 = \frac{\partial\rho_0(P)}{\partial P} dP$ , and then in (4), the specific energy  $\Pi$  will be a function of the pressure:  $\Pi = \Pi(P)$ .

On the other hand, we can assume that pressure is a function of mass density and is written as a barotropic relation:  $P = P(\rho_0)$ ,  $dP = \frac{\partial P(\rho_0)}{\partial \rho_0} d\rho_0$ . In this case, in (4), the specific energy  $\Pi$  is a function of the mass density:

$$\Pi = \Pi(\rho_0), \quad d\Pi = \frac{P(\rho_0)}{\rho_0^2} d\rho_0. \quad (5)$$

In (2), we can assume that

$$\mathcal{L} = \mathcal{L} \left[ g^{\mu\nu}, R(g^{\mu\nu}), x^\mu, \rho_{0q}(x^\mu, g^{\mu\nu}), \rho_0(x^\mu, g^{\mu\nu}), j^\beta(x^\mu, g^{\mu\nu}), A_\beta, F_{\alpha\beta}(A_\beta) \right], \quad (6)$$

that is, the Lagrangian density is a function of the independent variables  $x^\mu$ ,  $A_\beta$  and  $g^{\mu\nu}$ . In general, the scalar curvature  $R$  depends not only on the metric tensor  $g^{\mu\nu}$ , but also on the first and second derivatives of the metric tensor with respect to the four-coordinates  $x^\mu$ . In (6), the notation  $R(g^{\mu\nu})$  means that in the procedure for varying the Lagrangian density  $\mathcal{L}$ , the scalar curvature  $R$  should be varied not with respect to  $x^\mu$ , but with respect to  $g^{\mu\nu}$ . As for the quantities  $\rho_{0q}$ ,  $\rho_0$ ,  $j^\beta$  in (6), the variations of these quantities depend on  $x^\mu$  and  $g^{\mu\nu}$ .

We substitute (2) into (1) and, according to the principle of least action, set the variation  $\delta_g S$  of the action function with respect to the metric tensor  $g^{\mu\nu}$  to zero. For the cosmological constant, the variation is  $\delta_g \Lambda = 0$ . Since the tensor  $F_{\alpha\beta} = \nabla_\alpha A_\beta - \nabla_\beta A_\alpha$  depends only on the four-potential  $A_\beta$ , but not on  $g^{\mu\nu}$ , the variation of  $F_{\alpha\beta}$  with respect to the metric tensor behaves like a constant. This is due to the rule of calculating the variation with respect to a variable, which states that the variation with respect to that variable from other independent variables or their functions is equal to zero. Taking this into account, we find the variation of the action function with respect to the metric tensor  $g^{\mu\nu}$ :

$$\begin{aligned} \delta_g S &= \frac{1}{c} \int \delta_g (\mathcal{L} \sqrt{-g}) dx^0 dx^1 dx^2 dx^3 = \\ &= \frac{1}{c} \int \delta_g \mathcal{L} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 + \frac{1}{c} \int \mathcal{L} \delta_g (\sqrt{-g}) dx^0 dx^1 dx^2 dx^3 = \\ &= \frac{1}{c} \int \left\{ ck \delta_g R - c^2 \delta_g \rho_0 - \Pi \delta_g \rho_0 - \rho_0 \delta_g \Pi - \right. \\ &\quad \left. - A_\beta \delta_g j^\beta - \frac{1}{4\mu_0} F_{\alpha\beta} F_{\lambda\eta} \delta_g (g^{\lambda\alpha} g^{\eta\beta}) \right\} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 + \\ &\quad + \frac{1}{c} \int \mathcal{L} \delta_g (\sqrt{-g}) dx^0 dx^1 dx^2 dx^3 = 0. \end{aligned} \quad (7)$$

In (7), the relation  $F_{\alpha\beta}F^{\alpha\beta} = F_{\alpha\beta}F_{\lambda\eta}g^{\lambda\alpha}g^{\eta\beta}$  was used.

In [6], [8], the following expressions for variations on the metric tensor can be found, which are used in the least action principle:

$$\delta g^{\mu\nu} = -g^{\mu\alpha}g^{\nu\beta}\delta g_{\alpha\beta}, \quad \delta g_{\alpha\beta} = -g_{\alpha\mu}g_{\beta\nu}\delta g^{\mu\nu}.$$

$$\delta_g(\sqrt{-g}) = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu} = \frac{1}{2}\sqrt{-g}g^{\alpha\beta}\delta g_{\alpha\beta}. \quad \delta_g R = R_{\mu\nu}\delta g^{\mu\nu}. \quad (8)$$

In (8), the variations  $\delta g^{\mu\nu}$  and  $\delta g_{\alpha\beta}$  follow from the relation  $\delta(\delta^\nu_\alpha) = 0 = \delta(g_{\alpha\mu}g^{\mu\nu}) = g^{\nu\beta}\delta g_{\alpha\beta} + g_{\alpha\mu}\delta g^{\mu\nu}$ , where  $\delta^\nu_\alpha$  is the Kronecker symbol.

According to [6], the variation of mass density and the variation of charge density with respect to the metric tensor, taking into account (8), are equal to

$$\begin{aligned} \delta_g \rho_0 &= \frac{\rho_0}{2} \left( \frac{u^\alpha u^\beta}{c^2} - g^{\alpha\beta} \right) \delta g_{\alpha\beta} = -\frac{\rho_0}{2} \left( \frac{u^\alpha u^\beta}{c^2} - g^{\alpha\beta} \right) g_{\alpha\mu}g_{\beta\nu}\delta g^{\mu\nu} = \\ &= -\frac{\rho_0}{2} \left( \frac{u_\mu u_\nu}{c^2} - g_{\mu\nu} \right) \delta g^{\mu\nu}. \end{aligned} \quad (9)$$

$$\delta_g \rho_{0q} = \frac{\rho_{0q}}{2} \left( \frac{u^\alpha u^\beta}{c^2} - g^{\alpha\beta} \right) \delta g_{\alpha\beta} = -\frac{\rho_{0q}}{2} \left( \frac{u_\mu u_\nu}{c^2} - g_{\mu\nu} \right) \delta g^{\mu\nu} \quad (10)$$

The continuity equations for the mass four-current  $J^\sigma$  and the charge four-current  $j^\sigma$  are written as follows:

$$\nabla_\sigma J^\sigma = \nabla_\sigma (\rho_0 u^\sigma) = 0, \quad \nabla_\sigma j^\sigma = \nabla_\sigma (\rho_{0q} u^\sigma) = 0, \quad (11)$$

where the variations of the four-currents with respect to the metric tensor are as follows:

$$\delta_g J^\beta = \frac{1}{2} J^\beta g_{\mu\nu} \delta g^{\mu\nu}, \quad \delta_g j^\beta = \frac{1}{2} j^\beta g_{\mu\nu} \delta g^{\mu\nu}. \quad (12)$$

The relations are also fulfilled

$$\delta_g \left( J^\alpha \sqrt{-g} \right) = 0, \quad \delta_g \left( j^\alpha \sqrt{-g} \right) = 0. \quad (13)$$

By replacing the differentials  $d\Pi$  and  $d\rho_0$  in (5) with the corresponding variations  $\delta\Pi$  and  $\delta\rho_0$ , and using (9), we find the variation  $\delta\Pi$ :

$$\delta_g \Pi = \frac{P}{\rho_0^2} \delta_g \rho_0 = -\frac{P}{2\rho_0} \left( \frac{u_\mu u_\nu}{c^2} - g_{\mu\nu} \right) \delta g^{\mu\nu}. \quad (14)$$

The variation in the expression  $F_{\alpha\beta} F_{\lambda\eta} \delta(g^{\lambda\alpha} g^{\eta\beta})$ , taking into account the symmetry of the indices of the metric tensor  $g^{\mu\nu}$  and the antisymmetry of the indices of the tensor  $F_{\alpha\beta}$ , is obtained in the following form:

$$\begin{aligned} F_{\alpha\beta} F_{\lambda\eta} \delta_g (g^{\lambda\alpha} g^{\eta\beta}) &= F_{\alpha\beta} F_{\lambda\eta} g^{\lambda\alpha} \delta g^{\eta\beta} + F_{\alpha\beta} F_{\lambda\eta} g^{\eta\beta} \delta g^{\lambda\alpha} = \\ &= (F_{\alpha\nu} F_{\lambda\mu} g^{\lambda\alpha} + F_{\nu\beta} F_{\mu\eta} g^{\eta\beta}) \delta g^{\mu\nu} = \\ &= (F_{\alpha\nu} F_{\lambda\mu} g^{\lambda\alpha} + F_{\nu\alpha} F_{\mu\lambda} g^{\lambda\alpha}) \delta g^{\mu\nu} = 2F_{\alpha\nu} F_{\lambda\mu} g^{\lambda\alpha} \delta g^{\mu\nu}. \end{aligned} \quad (15)$$

Substituting (2), (8-9), (12), (14-15) into (7) gives:

$$\delta_g S = \frac{1}{c} \int \left\{ \begin{aligned} &ckR_{\mu\nu} - \frac{1}{2}ckRg_{\mu\nu} + ck\Lambda g_{\mu\nu} + \\ &+ \frac{1}{2}\rho_0 u_\mu u_\nu + \frac{1}{2c^2}\rho_0 \Pi u_\mu u_\nu + \\ &+ \frac{1}{2c^2}Pu_\mu u_\nu - \frac{1}{2}Pg_{\mu\nu} - \frac{1}{2\mu_0}F_{\alpha\nu}F_{\lambda\mu}g^{\lambda\alpha} + \\ &+ \frac{1}{8\mu_0}F_{\alpha\beta}F^{\alpha\beta}g_{\mu\nu} \end{aligned} \right\} \delta g^{\mu\nu} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = 0. \quad (16)$$

The variation  $\delta_g S$  of the action is zero if the square bracket in (16) is zero. This leads to the equation

$$ck \left( R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} \right) = -\frac{1}{2} \rho_0 u_\mu u_\nu - \frac{1}{2c^2} \rho_0 \Pi u_\mu u_\nu - \frac{1}{2c^2} P u_\mu u_\nu + \\ + \frac{1}{2} P g_{\mu\nu} + \frac{1}{2\mu_0} F_{\alpha\nu} F_{\lambda\mu} g^{\lambda\alpha} - \frac{1}{8\mu_0} F_{\alpha\beta} F^{\alpha\beta} g_{\mu\nu}. \quad (17)$$

The stress-energy tensor  $\tau_{\mu\nu}$  of an elastic fluid and the stress-energy tensor  $W_{\mu\nu}$  of electromagnetic field are defined by the following equations:

$$\tau_{\mu\nu} = \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u_\mu u_\nu - P g_{\mu\nu}. \\ W_{\mu\nu} = \frac{1}{\mu_0} \left( -F_{\alpha\nu} F_{\lambda\mu} g^{\lambda\alpha} + \frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} g_{\mu\nu} \right) = \frac{1}{\mu_0} \left( F_{\alpha\nu} F_{\mu\lambda} g^{\lambda\alpha} + \frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} g_{\mu\nu} \right) = \\ = \frac{1}{\mu_0} \left( F_{\mu\lambda} F^\lambda{}_\nu + \frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} g_{\mu\nu} \right). \quad (18)$$

Substituting (18) into (17) and taking into account (3), we find:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G\beta}{c^4} (\tau_{\mu\nu} + W_{\mu\nu}) = \frac{8\pi G\beta}{c^4} T_{\mu\nu}. \quad (19)$$

Equation (19) for determining the components of the metric tensor contains on the right-hand side the stress-energy tensor  $T_{\mu\nu} = \tau_{\mu\nu} + W_{\mu\nu}$  of the physical system.

After multiplying all the terms in (19) by  $g^{\mu\nu}$  and taking into account (18) and the expressions

$$R_{\mu\nu} g^{\mu\nu} = R, \quad g_{\mu\nu} g^{\mu\nu} = 4, \quad \tau_{\mu\nu} g^{\mu\nu} = c^2 \rho_0 + \rho_0 \Pi - 3P, \quad W_{\mu\nu} g^{\mu\nu} = 0, \quad (20)$$

we obtain the following:

$$-R + 4\Lambda = \frac{8\pi G\beta}{c^4} (c^2 \rho_0 + \rho_0 \Pi - 3P). \quad (21)$$

Expressing the cosmological constant  $\Lambda$  from (21) and substituting it into (19), we find:

$$R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu} = \frac{8\pi G\beta}{c^4} \left( \tau_{\mu\nu} + W_{\mu\nu} - \frac{c^2}{4} \rho_0 g_{\mu\nu} - \frac{1}{4} \rho_0 \Pi g_{\mu\nu} + \frac{3}{4} P g_{\mu\nu} \right). \quad (22)$$

Equation (22) allows us to find the components of the metric tensor both inside and outside the fluid, within the framework of general theory of relativity. Since (22) does not contain the cosmological constant  $\Lambda$ , which is not known in advance, equation (22) is more convenient than equation (19).

Let us apply the covariant derivative  $\nabla^\nu$  to all terms in equation (19), taking into account the expressions for the stress-energy tensors (18). The action of the covariant derivative  $\nabla^\nu$  on the left side of (19) gives zero due to the property of the Einstein tensor  $\nabla^\nu \left( R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) = 0$  and equality  $\nabla^\nu (\Lambda g_{\mu\nu}) = \nabla_\mu \Lambda = 0$  for the cosmological constant  $\Lambda$ . With this in mind, for the right-hand side of (19) we have:

$$\nabla_\sigma \left[ \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u_\mu u^\sigma \right] - \nabla_\mu P + j^\sigma F_{\sigma\mu} = 0. \quad (23)$$

In (23), it was taken into account that  $\nabla^\nu W_{\mu\nu} = j^\sigma F_{\sigma\mu}$ , as a consequence of the electromagnetic field equations, and in the first term in the covariant derivative  $\nabla^\nu$ , the transition to the derivative  $\nabla_\sigma$  was made simultaneously with the transition from  $u_\nu$  to  $u^\sigma$ .

Outside the fluid, all terms in (23) are equal to zero, and in the fluid, (23) represents the equation of motion for the fluid particles, which follows from the equation for the metric (19).

### 3. The balance equation for force densities



We substitute the Lagrangian density  $\mathcal{L}$  (2) into the action function  $S$  (1), calculate the variation  $\delta_x S$  associated with the variation  $\delta x^\mu$  of the coordinates, and set  $\delta_x S$  equal to zero. Note that  $\delta_x \Lambda = 0$ , the scalar curvature  $R$  and  $\sqrt{-g}$  depend on the  $g^{\mu\nu}$  and are independent on  $x^\mu$ , while the invariant quantity  $F_{\alpha\beta} F^{\alpha\beta} = F_{\alpha\beta} F_{\lambda\eta} g^{\lambda\alpha} g^{\eta\beta}$  depends on the four-potential  $A_\alpha$  and  $g^{\mu\nu}$ , and is independent on  $x^\mu$ . This leads to the following equation:

$$\delta_x S = \frac{1}{c} \int \left( -c^2 \delta_x \rho_0 - \rho_0 \delta_x \Pi - \Pi \delta_x \rho_0 - A_\beta \delta_x j^\beta \right) \sqrt{-g} dx^0 dx^1 dx^2 dx^3. \quad (24)$$

According to [6], the variations of four-currents and mass density on four-coordinates are equal to:

$$\delta_x J^\beta = \nabla_\sigma \left( J^\sigma \delta x^\beta - J^\beta \delta x^\sigma \right) = \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\sigma} \left[ \sqrt{-g} \left( J^\sigma \delta x^\beta - J^\beta \delta x^\sigma \right) \right].$$

$$\delta_x j^\beta = \nabla_\sigma \left( j^\sigma \delta x^\beta - j^\beta \delta x^\sigma \right) = \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\sigma} \left[ \sqrt{-g} \left( j^\sigma \delta x^\beta - j^\beta \delta x^\sigma \right) \right].$$

$$\delta_x \rho_0 = -\nabla_\beta \left( \rho_0 \delta x^\beta \right) + \frac{\rho_0}{c^2} u_\beta u^\sigma \nabla_\sigma \left( \delta x^\beta \right). \quad (25)$$

Considering (5) and (25), for the variation  $\delta_x \Pi$  we find:

$$\delta_x \Pi = \frac{P}{\rho_0^2} \delta_x \rho_0 = -\frac{P}{\rho_0^2} \nabla_\beta \left( \rho_0 \delta x^\beta \right) + \frac{P}{c^2 \rho_0} u_\beta u^\sigma \nabla_\sigma \left( \delta x^\beta \right). \quad (26)$$

We substitute (25-26) in (24):

$$\delta_x S = \frac{1}{c} \int \left( \begin{aligned} & c^2 \nabla_\beta (\rho_0 \delta x^\beta) - \rho_0 u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) + \\ & + \frac{P}{\rho_0} \nabla_\beta (\rho_0 \delta x^\beta) - \frac{P}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) + \\ & + \Pi \nabla_\beta (\rho_0 \delta x^\beta) - \frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) - \\ & - A_\beta \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\sigma} [\sqrt{-g} (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma)] \end{aligned} \right) \sqrt{-g} dx^0 dx^1 dx^2 dx^3. \quad (27)$$

Variation  $\delta_x S$  (27) is calculated in Appendix A. As a result of this variation in (A22), the force density balance equation is obtained in the following form:

$$\nabla_\sigma \left[ \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u_\mu u^\sigma \right] - \nabla_\mu P + \frac{P}{\rho_0} \nabla_\mu \rho_0 - \rho_0 \nabla_\mu \Pi + j^\sigma F_{\sigma\mu} = 0. \quad (28)$$

Considering that  $\frac{d\Pi}{d\rho_0} = \frac{P}{\rho_0^2}$  in (4), and  $\Pi$  is a scalar function, we transform  $\nabla_\mu \Pi$  in the fourth term in (28):

$$\nabla_\mu \Pi = \frac{\partial \Pi}{\partial x^\mu} = \frac{\partial \Pi}{\partial \rho_0} \frac{\partial \rho_0}{\partial x^\mu} = \frac{P}{\rho_0^2} \frac{\partial \rho_0}{\partial x^\mu} = \frac{P}{\rho_0^2} \nabla_\mu \rho_0. \quad (29)$$

Substituting (29) into (28) gives:

$$\nabla_\sigma \left[ \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u_\mu u^\sigma \right] - \nabla_\mu P + j^\sigma F_{\sigma\mu} = 0. \quad (30)$$

The equation of motion (30), which appears as a result of variation of the action function with respect to the four-coordinates, coincides with the equation of motion (23), which is a consequence of variation of the action function with respect to the metric tensor. This demonstrates the self-consistency of the principle of least action.

Considering the continuity equation  $\nabla_\sigma (\rho_0 u^\sigma) = 0$  (11) for the mass four-current, the first term in (30) can be rewritten as follows:

$$\begin{aligned} \nabla_\sigma \left[ \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u_\mu u^\sigma \right] &= \left( 1 + \frac{\Pi}{c^2} + \frac{P}{c^2 \rho_0} \right) u_\mu \nabla_\sigma (\rho_0 u^\sigma) + \\ &+ \rho_0 u^\sigma \nabla_\sigma \left[ \left( 1 + \frac{\Pi}{c^2} + \frac{P}{c^2 \rho_0} \right) u_\mu \right] = \rho_0 u^\sigma \nabla_\sigma \left[ \left( 1 + \frac{\Pi}{c^2} + \frac{P}{c^2 \rho_0} \right) u_\mu \right]. \end{aligned} \quad (31)$$

Using the expression for the operator of proper- time-derivative  $u^\sigma \nabla_\sigma = \frac{D}{D\tau}$ , from (30-31) follows the equation of motion of matter particles, taking into account the density of the electromagnetic Lorentz force  $j^\sigma F_{\mu\sigma} = -j^\sigma F_{\sigma\mu}$ :

$$\rho_0 u^\sigma \nabla_\sigma \left[ \left( 1 + \frac{\Pi}{c^2} + \frac{P}{c^2 \rho_0} \right) u_\mu \right] = \rho_0 \frac{D}{D\tau} \left[ \left( 1 + \frac{\Pi}{c^2} + \frac{P}{c^2 \rho_0} \right) u_\mu \right] = \nabla_\mu P + F_{\mu\sigma} j^\sigma. \quad (32)$$

According to (5), the specific elastic compression energy  $\Pi$  of the fluid and the scalar pressure are some functions of the invariant mass density  $\rho_0$ . In the case where the mass density  $\rho_0$  does not depend on the pressure  $P$ , the specific energy  $\Pi$  in (4) becomes zero. Then the equation of motion (32) ceases to depend on  $\Pi$ , and the tensor  $\tau_{\mu\nu}$  (18) becomes the stress-energy tensor of an ideal fluid, in which the energy of fluid compression is not taken into account.

The left-hand side of (32) can be further transformed by noting that the covariant derivative  $\nabla_\sigma$  acts on the product of two functions:

$$\rho_0 u^\sigma \nabla_\sigma \left[ \left( 1 + \frac{\Pi}{c^2} + \frac{P}{c^2 \rho_0} \right) u_\mu \right] = \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u^\sigma \nabla_\sigma u_\mu + \frac{1}{c^2} \rho_0 u_\mu u^\sigma \nabla_\sigma \left( \Pi + \frac{P}{\rho_0} \right). \quad (33)$$

For the last term on the right-hand side of (33), taking into account  $\nabla_\mu \Pi$  (29), we find:

$$\begin{aligned} \frac{1}{c^2} \rho_0 u_\mu u^\sigma \nabla_\sigma \left( \Pi + \frac{P}{\rho_0} \right) &= \frac{1}{c^2} \rho_0 u_\mu u^\sigma \nabla_\sigma \Pi + \frac{1}{c^2} \rho_0 u_\mu u^\sigma \nabla_\sigma \left( \frac{P}{\rho_0} \right) = \\ &= \frac{1}{c^2} u_\mu u^\sigma \frac{P}{\rho_0} \nabla_\sigma \rho_0 + \frac{1}{c^2} u_\mu u^\sigma \rho_0 \nabla_\sigma \left( \frac{P}{\rho_0} \right) = \frac{1}{c^2} u_\mu u^\sigma \nabla_\sigma P. \end{aligned} \quad (34)$$

From (33-34), the equation follows:

$$\rho_0 u^\sigma \nabla_\sigma \left[ \left( 1 + \frac{\Pi}{c^2} + \frac{P}{c^2 \rho_0} \right) u_\mu \right] = \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u^\sigma \nabla_\sigma u_\mu + \frac{1}{c^2} u_\mu u^\sigma \nabla_\sigma P. \quad (35)$$

Substituting (35) into (32) and taking the operator of proper- time-derivative leads to the equation of motion

$$\left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) \frac{Du_\mu}{D\tau} = \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) A_\mu = \nabla_\mu P - \frac{1}{c^2} u_\mu u^\sigma \nabla_\sigma P + F_{\mu\sigma} j^\sigma. \quad (36)$$

The right-hand side of (36) contains the density of the pressure force  $\nabla_\mu P$  and the density of the electromagnetic Lorentz force  $F_{\mu\sigma} j^\sigma$ , under the influence of which the four-acceleration  $A_\mu = \frac{Du_\mu}{D\tau}$  of the substance particles occurs. Since the four-velocity with a covariant index is expressed by the formula  $u_\mu = g_{\mu\nu} u^\nu$ , the components of the metric tensor  $g_{\mu\nu}$  appear both on the left and on the right sides of (36). Thus, the gravitational field, which is replaced by the metric field in the general theory of relativity, appears in (36) through the metric tensor and its derivatives with respect to four-coordinates, and is taken into account in the equation of motion.

#### 4. Scalar curvature outside of matter

We consider a stationary massive charged spherical body, the matter of which is a certain fluid. Outside of such body, according to the general theory of relativity, there is only an electromagnetic field, which is described by the stress-energy tensor  $W_{\mu\nu}$ . In this case, the equation for the metric (22) is written as follows:

$$R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu} = \frac{8\pi G \beta}{c^4} W_{\mu\nu}. \quad (37)$$

The solution of equation (37) is presented in Appendix B, where, according to (B26) and (B27), the nonzero components of the metric tensor and the scalar curvature  $R$  are found:

$$g_{00} = B = 1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \varepsilon_0 c^4 r^2}, \quad g_{11} = -K = -\frac{1}{1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \varepsilon_0 c^4 r^2}}.$$

$$g_{22} = -E = -r^2, \quad g_{33} = -E \sin^2 \theta = -r^2 \sin^2 \theta. \quad (38)$$

$$R = 0. \quad (39)$$

In (38),  $C_2$  is a constant,  $Q$  is the total electric charge of the body.  $G$  and  $\varepsilon_0$  denote the gravitational and electric constants.

There is no matter outside the body,  $\rho_0 = 0$ ,  $\Pi = 0$ ,  $P = 0$ . Then in (21), the relation  $-R + 4\Lambda = 0$  is satisfied, and the scalar curvature  $R$  in (39) is equal to zero. It follows that the cosmological constant outside the body is also equal to zero:

$$\Lambda = 0. \quad (40)$$

## 5. Energy

In [9], the main problems of the general theory of relativity associated with the definition of gravitational energy in terms of various pseudotensors are described. Gravitational energy is considered as a quasi-local quantity, which is most accurately determined only for the entire system if the metric becomes asymptotically flat at infinity.

On the other hand, the relativistic energy density, which determines the inertial mass density of a system, is proportional to the time component of the system's four-momentum density. In this case, a given gravitational energy pseudotensor can be used to determine the gravitational force density, but not to calculate the gravitational energy density. This is due to the fact that the time components of the gravitational energy pseudotensor do not form a four-vector, so the integral over the volume of these components cannot be a four-vector contributing to the relativistic energy. This circumstance is pointed out in [10]. An example here is the energy-momentum tensor of electromagnetic field, the divergence of which yields the electromagnetic four-force density and contributes to the generalized Poynting theorem [11]. However, the time components of this tensor do not form the electromagnetic four-momentum density, as indicated by the well-known 4/3 problem.

In this regard, to determine the energy of the system under consideration, we will assume, as in [12-13], that the Lagrangian density depends only on the coordinate time  $t$ , the three-dimensional position vectors  $\mathbf{r} = (x^1, x^2, x^3)$  of particles, and the velocities  $\mathbf{v} = \frac{d\mathbf{r}}{dt} = \left( \frac{dx^1}{dt}, \frac{dx^2}{dt}, \frac{dx^3}{dt} \right)$  of the particles, in the form  $\mathcal{L} = \mathcal{L}(t, \mathbf{r}, \mathbf{v})$ . Given this in (1), we have for the action function:

$$S = \frac{1}{c} \int \mathcal{L}(t, \mathbf{r}, \mathbf{v}) \sqrt{-g} dx^0 dx^1 dx^2 dx^3. \quad (41)$$

Previously, based on the action function (41), in [10] the energy and momentum of a moving system were determined, and the four-momentum of the system as the sum of generalized four-momentum of particles and the four-momentum of the system's fields was calculated.

The variation of the action function with respect to the particle vectors  $\mathbf{r}$ , which we denote  $\delta_{\mathbf{r}} S$ , is equal to:

$$\begin{aligned} \delta_{\mathbf{r}} S &= \frac{1}{c} \int \left[ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{r}} \cdot \delta \mathbf{r} + \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \cdot \delta \mathbf{v} \right] dx^0 dx^1 dx^2 dx^3 = \\ &= \frac{1}{c} \int \left[ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{r}} \cdot \delta \mathbf{r} + \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \cdot \frac{d\delta \mathbf{r}}{dt} \right] dx^0 dx^1 dx^2 dx^3 = \\ &= \frac{1}{c} \int \left\{ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{r}} \cdot \delta \mathbf{r} + \frac{d}{dt} \left[ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \cdot \delta \mathbf{r} \right] - \delta \mathbf{r} \cdot \frac{d}{dt} \left[ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \right] \right\} dx^0 dx^1 dx^2 dx^3. \end{aligned} \quad (42)$$

In (42), the relations  $\mathbf{v} = \frac{d\mathbf{r}}{dt}$ ,  $\delta \mathbf{v} = \delta \frac{d\mathbf{r}}{dt} = \frac{d\delta \mathbf{r}}{dt}$  have been taken into account. For the second term on the right-hand side of (42), taking into account  $x^0 = ct$ , we find:

$$\begin{aligned}
& \frac{1}{c} \int \frac{d}{dt} \left[ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \cdot \delta \mathbf{r} \right] dx^0 dx^1 dx^2 dx^3 = \int_t \frac{d}{dt} \left[ \int_X \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \cdot \delta \mathbf{r} dx^1 dx^2 dx^3 \right] dt = \\
& = \left[ \int_X \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \cdot \delta \mathbf{r} dx^1 dx^2 dx^3 \right]_{t_2} - \left[ \int_X \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \cdot \delta \mathbf{r} dx^1 dx^2 dx^3 \right]_{t_1} = 0.
\end{aligned} \tag{43}$$

The equality to zero in (43) follows from the variation condition, according to which the variations  $\delta \mathbf{r}$  at the initial time  $t_1$  and at the final time  $t_2$  are equal to zero. Taking into account (43), the expression (42) is simplified:

$$\delta_r S = \frac{1}{c} \int \left\{ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{r}} - \frac{d}{dt} \left[ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \right] \right\} \cdot \delta \mathbf{r} dx^0 dx^1 dx^2 dx^3. \tag{44}$$

Equating the variation  $\delta_r S$  to zero, given that the variation  $\delta \mathbf{r}$  is not equal to zero, we obtain the following relation:

$$\frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{r}} = \frac{d}{dt} \left[ \frac{\partial(\mathcal{L}\sqrt{-g})}{\partial \mathbf{v}} \right]. \tag{45}$$

According to [8], the following relation holds:

$$\sqrt{-g} = \sqrt{\gamma} \sqrt{g_{00}}, \tag{46}$$

where  $g_{00}$  is the time component of the four-dimensional metric tensor,  $\gamma$  is the determinant of the three-dimensional metric tensor  $\gamma_{\alpha\beta} = -g_{\alpha\beta} + \frac{g_{0\alpha}g_{0\beta}}{g_{00}}$ , the indices  $\alpha$  and  $\beta$  run over the values 1,2,3.

Substituting (46) into (45) gives the following:

$$\frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{r}} = \frac{d}{dt} \left[ \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \right] + \frac{1}{\sqrt{g_{00}}} \frac{d\sqrt{g_{00}}}{dt} \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} + \frac{1}{\sqrt{g_{00}}} \frac{d}{dt} \left( \mathcal{L}\sqrt{\gamma} \frac{\partial\sqrt{g_{00}}}{\partial \mathbf{v}} \right) - \frac{\mathcal{L}\sqrt{\gamma}}{\sqrt{g_{00}}} \frac{\partial\sqrt{g_{00}}}{\partial \mathbf{r}}. \quad (47)$$

We define the Lagrangian function  $L$  by the following expression:

$$L = \int \mathcal{L} \sqrt{\gamma} dx^1 dx^2 dx^3 = \int \mathcal{L}_\tau (dx^1 dx^2 dx^3)_\tau = L_\tau. \quad (48)$$

The quantity  $(dx^1 dx^2 dx^3)_\tau$  in (48) is the element of the three-dimensional volume from the point of view of a stationary local observer in gravitational field and counting time  $\tau$ . For the coordinate observer, the invariant element of the three-dimensional volume is  $\sqrt{\gamma} dx^1 dx^2 dx^3$ , with  $(dx^1 dx^2 dx^3)_\tau = \sqrt{\gamma} dx^1 dx^2 dx^3$ . The Lagrangian density  $\mathcal{L}$  in (48) is expressed in terms of the coordinates  $x^\mu = x^0, x^1, x^2, x^3$ , and the Lagrangian density  $\mathcal{L}_\tau$  and the Lagrangian function  $L_\tau$  are expressed in terms of the coordinates  $(x^\mu)_\tau = (x^0, x^1, x^2, x^3)_\tau$  used by the local observers.

Given (46), (48), and the relations  $dx^0 = c dt$ ,  $d\tau = \sqrt{g_{00}} dt$ , we find for the action in (1):

$$\begin{aligned} S &= \frac{1}{c} \int \mathcal{L} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \int \mathcal{L} \sqrt{\gamma} \sqrt{g_{00}} dt dx^1 dx^2 dx^3 = \int L \sqrt{g_{00}} dt = \\ &= \int L_\tau d\tau = \int \mathcal{L}_\tau d\tau (dx^1 dx^2 dx^3)_\tau = \int \mathcal{L}_0 d\tau_0 (dx^1 dx^2 dx^3)_0 = \int L_0 d\tau_0. \end{aligned} \quad (49)$$

In (49),  $L_0 = \int \mathcal{L}_0 (dx^1 dx^2 dx^3)_0$  is the sum of the Lagrangian functions for the matter elements, each of which has the proper time  $\tau_0$  and moves relative to the nearby corresponding stationary local observer, which measures time  $\tau$ . In (49), the equality  $d\tau (dx^1 dx^2 dx^3)_\tau = d\tau_0 (dx^1 dx^2 dx^3)_0$  of the four-dimensional volume elements is assumed, which is possible since the observer accompanying the matter element and the local observer are at the same point in spacetime and can check their readings. The peculiarity of (48-49) is that the action function  $S$ , the Lagrangian  $L$  and the Lagrangian density  $\mathcal{L}$  are scalar invariant functions whose values do not depend on the choice of four-dimensional coordinates.



The dependence of the Lagrangian density  $\mathcal{L}(t, \mathbf{r}, \mathbf{v})$  on the position  $\mathbf{r}$  and velocity  $\mathbf{v}$  of the particles in (49) is given from the point of view of coordinate observer. For a local observer past whom a particle is moving, the Lagrangian density  $\mathcal{L}_\tau = \mathcal{L}_\tau(\tau, \mathbf{r}_\tau, \mathbf{v}_\tau)$  depends on the particle's velocity  $\mathbf{v}_\tau$ , which can be defined similarly to [8] in the form  $\mathbf{v}_\tau = \frac{dl}{d\tau}$ , where  $dl$  is the differential of physical displacement.

We calculate the differential of the Lagrangian function in (48) taking into account that the Lagrangian density  $\mathcal{L}$  and  $\sqrt{\gamma}$  can depend on the coordinate time  $t$ , on the three-dimensional position vectors  $\mathbf{r}$  of particles and on the velocities  $\mathbf{v}$  of motion of these particles. Using  $\frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{r}}$  (47), we find:

$$\begin{aligned}
dL &= \int d(\mathcal{L}\sqrt{\gamma}) dx^1 dx^2 dx^3 = \\
&= \int \left[ \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial t} dt + \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{r}} \cdot d\mathbf{r} + \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \cdot d\mathbf{v} \right] dx^1 dx^2 dx^3 = \\
&= \int \left[ \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial t} dt + d\mathbf{r} \cdot \frac{d}{dt} \left[ \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \right] + \frac{1}{\sqrt{g_{00}}} \frac{d\sqrt{g_{00}}}{dt} \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \cdot d\mathbf{r} + \right. \\
&\quad \left. + \frac{1}{\sqrt{g_{00}}} d\mathbf{r} \cdot \frac{d}{dt} \left[ \mathcal{L}\sqrt{\gamma} \frac{\partial \sqrt{g_{00}}}{\partial \mathbf{v}} \right] - \frac{\mathcal{L}\sqrt{\gamma}}{\sqrt{g_{00}}} \frac{\partial \sqrt{g_{00}}}{\partial \mathbf{r}} \cdot d\mathbf{r} + \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \cdot d\mathbf{v} \right] dx^1 dx^2 dx^3.
\end{aligned} \tag{50}$$

Let's divide all the terms in (50) by  $dt$  and make a substitution in the form of  $\mathbf{v} = \frac{d\mathbf{r}}{dt}$ :

$$\begin{aligned}
\frac{dL}{dt} &= \int \left[ \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial t} + \frac{d}{dt} \left[ \mathbf{v} \cdot \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \right] + \frac{1}{\sqrt{g_{00}}} \frac{d\sqrt{g_{00}}}{dt} \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \cdot \mathbf{v} + \right. \\
&\quad \left. + \frac{1}{\sqrt{g_{00}}} \mathbf{v} \cdot \frac{d}{dt} \left[ \mathcal{L}\sqrt{\gamma} \frac{\partial \sqrt{g_{00}}}{\partial \mathbf{v}} \right] - \frac{\mathcal{L}\sqrt{\gamma}}{\sqrt{g_{00}}} \frac{\partial \sqrt{g_{00}}}{\partial \mathbf{r}} \cdot \mathbf{v} \right] dx^1 dx^2 dx^3.
\end{aligned} \tag{51}$$

From (51), we find the following:

$$\begin{aligned}
& \frac{d}{dt} \left\{ \int \left[ \mathbf{v} \cdot \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \right] dx^1 dx^2 dx^3 - L \right\} = \\
& = - \int \left[ \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial t} + \frac{1}{\sqrt{g_{00}}} \frac{d\sqrt{g_{00}}}{dt} \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \cdot \mathbf{v} + \right. \\
& \quad \left. + \frac{1}{\sqrt{g_{00}}} \mathbf{v} \cdot \frac{d}{dt} \left[ \mathcal{L}\sqrt{\gamma} \frac{\partial \sqrt{g_{00}}}{\partial \mathbf{v}} \right] - \frac{\mathcal{L}\sqrt{\gamma}}{\sqrt{g_{00}}} \frac{\partial \sqrt{g_{00}}}{\partial \mathbf{r}} \cdot \mathbf{v} \right] dx^1 dx^2 dx^3.
\end{aligned} \tag{52}$$

If the right-hand side of (52) is equal to zero, then the system conserves energy, which is defined as the function under the time derivative on the left-hand side of (52). Given the expression  $L$  (48), we can write:

$$E = \int \left[ \mathbf{v} \cdot \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} \right] dx^1 dx^2 dx^3 - L = \int \left[ \mathbf{v} \cdot \frac{\partial(\mathcal{L}\sqrt{\gamma})}{\partial \mathbf{v}} - \mathcal{L}\sqrt{\gamma} \right] dx^1 dx^2 dx^3. \tag{53}$$

To determine in spherical coordinates the determinant  $\gamma$  of the three-dimensional metric tensor  $\gamma_{\alpha\beta} = -g_{\alpha\beta} + \frac{g_{0\alpha}g_{0\beta}}{g_{00}}$ , where the indices  $\alpha$  and  $\beta$  run through the values 1,2,3, we use the expression for the four-dimensional metric tensor  $g_{\mu\nu}$  in (B3) in Appendix B:

$$g_{\mu\nu} = \begin{pmatrix} B & 0 & 0 & 0 \\ 0 & -K & 0 & 0 \\ 0 & 0 & -E & 0 \\ 0 & 0 & 0 & -E \sin^2 \theta \end{pmatrix}, \tag{54}$$

Since in (54)  $g_{0\beta} = 0$ , then for  $\gamma_{\alpha\beta}$  we have

$$\gamma_{\alpha\beta} = -g_{\alpha\beta} = \begin{pmatrix} K & 0 & 0 \\ 0 & E & 0 \\ 0 & 0 & E \sin^2 \theta \end{pmatrix}. \quad (55)$$

From (55), taking into account the ratio  $K = \frac{1}{B}$  and components of the metric tensor (38), we determine  $\sqrt{\gamma}$  outside the stationary spherical massive charged body:

$$\sqrt{\gamma} = \sqrt{KE^2 \sin^2 \theta} = \frac{E \sin \theta}{\sqrt{B}} = \frac{r^2 \sin \theta}{\sqrt{1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \varepsilon_0 c^4 r^2}}}. \quad (56)$$

Since  $\sqrt{\gamma}$  (56) does not depend on the velocity  $\mathbf{v}$ , the quantity  $\sqrt{\gamma}$  can be taken outside the square brackets in (53). Substituting (2) into (53), we find:

$$\begin{aligned} E &= \int \left( \mathbf{v} \cdot \frac{\partial \mathcal{L}}{\partial \mathbf{v}} - \mathcal{L} \right) \sqrt{\gamma} dx^1 dx^2 dx^3 = \\ &= \int \left[ \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} \left( ckR - 2ck\Lambda - c^2 \rho_0 - \rho_0 \Pi - A_\beta j^\beta - \frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} \right) - \right. \\ &\quad \left. -ckR + 2ck\Lambda + c^2 \rho_0 + \rho_0 \Pi + A_\beta j^\beta + \frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} \right] \sqrt{\gamma} dx^1 dx^2 dx^3. \end{aligned} \quad (57)$$

Let's consider the contribution of the quantity  $A_\beta j^\beta$  to the system's energy. Since

$$A_\beta = \left( \frac{\varphi}{c}, -\mathbf{A} \right), \quad j^\beta = \rho_{0q} u^\beta = \left( \rho_{0q} c \frac{dt}{d\tau}, \rho_{0q} \frac{dt}{d\tau} \mathbf{v} \right).$$

$$A_\beta j^\beta = \rho_{0q} \varphi \frac{dt}{d\tau} - \rho_{0q} \frac{dt}{d\tau} \mathbf{A} \cdot \mathbf{v},$$

then, as a first approximation, we can write:

$$\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} (-A_\beta j^\beta) = -\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} \left( \rho_{0q} \varphi \frac{dt}{d\tau} - \rho_{0q} \frac{dt}{d\tau} \mathbf{A} \cdot \mathbf{v} \right) \approx \rho_{0q} \frac{dt}{d\tau} \mathbf{A} \cdot \mathbf{v}.$$

$$\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} (-A_\beta j^\beta) + A_\beta j^\beta \approx \rho_{0q} \varphi \frac{dt}{d\tau}. \quad (58)$$

In (58),  $\tau$  is the proper time of a moving charged element of matter,  $t$  is the coordinate time,  $\varphi$  and  $\mathbf{A}$  are the global scalar and vector potentials of electromagnetic field at the location of the element of matter. It is assumed that  $\varphi$ ,  $\mathbf{A}$ ,  $\frac{dt}{d\tau}$ , and the charge density  $\rho_{0q}$  depend very weakly on the velocity  $\mathbf{v}$  of the charged element of matter under consideration. The quantity  $\frac{dt}{d\tau}$  in (58), in the limit of special relativity, represents the Lorentz factor  $\gamma$ , with

$$\frac{dt}{d\tau} \approx \gamma = \frac{1}{\sqrt{1 - \frac{\mathbf{v}^2}{c^2}}}.$$

Substituting (58) into (57) yields:

$$E \approx \int \left[ \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} \left( ckR - 2ck\Lambda - c^2\rho_0 - \rho_0\Pi - \frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} \right) - \right. \\ \left. -ckR + 2ck\Lambda + c^2\rho_0 + \rho_0\Pi + \rho_{0q}\varphi \frac{dt}{d\tau} + \frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} \right] \sqrt{\gamma} dx^1 dx^2 dx^3. \quad (59)$$

For a stationary body with chaotic motion of matter particles, to a first approximation, we can assume that the scalar curvature  $R$ , the cosmological constant  $\Lambda$ , the mass density  $\rho_0$ , and the tensor  $F_{\alpha\beta}$  depend weakly on the velocity  $\mathbf{v}$  of the matter particles, and therefore the derivative  $\frac{\partial}{\partial \mathbf{v}}$  of these quantities can be neglected. This yields the following in (59):

$$E \approx \int \left( -ckR + 2ck\Lambda + c^2\rho_0 + \rho_0\Pi - \rho_0 \mathbf{v} \cdot \frac{\partial \Pi}{\partial \mathbf{v}} + \rho_{0q}\varphi \frac{dt}{d\tau} + \frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} \right) \sqrt{\gamma} dx^1 dx^2 dx^3. \quad (60)$$

In order to calculate the energy inside the body in (60), we must first define the quantities  $R$ ,  $\Lambda$ ,  $\rho_0$ ,  $\Pi$ ,  $\mathbf{v}$ ,  $\rho_{0q}$ ,  $\varphi$ ,  $F_{\alpha\beta}$  as certain functions of coordinates and time. For our purposes, we will solve a simpler problem and calculate in (60) only the energy outside the body, where there is no matter and the mass density  $\rho_0 = 0$ , charge density  $\rho_{0q} = 0$ . We will take into account that outside the body  $R = 0$  in (39) and  $\Lambda = 0$  in (40), and use  $\sqrt{\gamma}$  (56):

$$E_o \approx \int \left( \frac{1}{4\mu_0} F_{\alpha\beta} F^{\alpha\beta} \right) \frac{r^2 \sin \theta}{\sqrt{1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \varepsilon_0 c^4 r^2}}} dr d\theta d\phi. \quad (61)$$

Given the relation  $K = \frac{1}{B}$  in (38), the components of the electromagnetic field tensor  $F_{\mu\nu}$  in (B2) and  $F^{\lambda\eta}$  in (B5) in Appendix B, we find:

$$F_{\alpha\beta} F^{\alpha\beta} = F_{01} F^{01} + F_{10} F^{10} = -\frac{Q^2}{8\pi^2 \varepsilon_0^2 c^2 r^4}. \quad (62)$$

As can be seen in (38), the component  $g_{00}$  of the metric tensor outside a stationary charged spherical body is equal to  $g_{00} = 1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \varepsilon_0 c^4 r^2}$ . This expression can be compared with the component  $g_{00}$  known in the Reissner–Nordstrom metric solution [14], and equal to  $g_{00} = 1 - \frac{2GM}{c^2 r} + \frac{Q^2 G \beta}{4\pi \varepsilon_0 c^4 r^2}$ , where  $M$  is the gravitational mass of the body,  $Q$  is the electric charge of the body,  $G$  and  $\varepsilon_0$  are the gravitational and electric constants. The constant  $C_2$  should be of the order of magnitude  $-\frac{2GM}{c^2}$ , and in most cases the ratio  $\frac{C_2}{r}$  is noticeably less than one. Substituting (62) into (61) and neglecting the square root in (61), taking into account the ratio  $\mu_0 \varepsilon_0 c^2 = 1$ , we find:

$$E_o \approx - \int_{r=a}^{\infty} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{Q^2}{32\pi^2 \varepsilon_0 r^2} \sin \theta dr d\theta d\phi = -\frac{Q^2}{8\pi \varepsilon_0 a}. \quad (63)$$

According to (63), the main contribution to the energy outside the body is made by the electromagnetic field of the body, and this contribution has a negative sign. To understand this result, one must take into account the integral theorem of field energy [15], according to which, in the system under consideration, in the absence of electric currents, the electrostatic energy  $E_s$  of the body's charges is positive and twice as large as the absolute value of the field energy  $E_o$  (63). The sum of the energies  $E_s$  and  $E_o$  gives a positive energy  $E_e$  equal to the absolute value of the field energy  $E_o$ , so that  $tE_e$  is a contribution from the charges and the electromagnetic field to the relativistic energy of the system.

## 6. Discussion of the results

In (23) and (30), the equations of motion of matter particles are presented, derived in different ways from the principle of least action. Thus, the predictions of general theory of relativity (GTR) in the description of particle motion are confirmed quite reliably.

The situation changes when it is necessary to determine the energy of a physical system within the framework of GTR. As calculations in Sections 4 and 5 show, outside a massive charged spherical body, the scalar curvature  $R$  and the cosmological constant  $\Lambda$  become zero. In (63), only the electromagnetic energy contributes to the energy of the system outside the body, while the gravitational field does not contribute to the system's energy. This contradicts the fact that in Newtonian physics, the gravitational field, like the electromagnetic field, has a potential and potential energy. The absence of gravitational energy outside the body means that in GTR, gravitational energy is localized in a way that is significantly different from the way energy is localized in an electromagnetic field.

In GTR, there are several ways to assign some gravitational energy to the space outside a massive body. To do this, expressions with the dimension of energy density are constructed using the components of the metric tensor, and almost all of these expressions turn out to be pseudotensors of the gravitational field [1]. The existence of several such pseudotensors suggests that there is still a problem in GTR regarding the contribution of gravitational field energy to the system's energy and the method of defining gravitational energy.

The obvious reason for the problem with the localization of gravitational energy in GTR is that in GTR, the effects of gravitation are completely expressed through the metric tensor and its time and coordinate derivatives, and the gravitational field is replaced by the metric field. As a result of this replacement, the metric field not only curves spacetime, introducing a new

degree of freedom into the equations of physics, but also replaces the gravitational interaction between bodies.

One example of a successful solution to the energy problem of the gravitational field is the covariant theory of gravitation [13], [16]. In this theory, each known particle interaction is associated with a corresponding vector relativistic field [17]. The transition from scalar to vector fields significantly enriches the theory by introducing new variables in the form of vector potentials, and the inclusion of the relativistic velocity transformation rule in the equations of motion allows for the consideration of phenomena in moving reference frames [18-19].

The use of vector fields in the covariant theory of gravitation leads to the fact that each field yields its own contribution to the equation of motion and to the energy of the system. Another advantage of the covariant theory of gravitation is that it makes it possible to present formulas and calculate the energy and momentum of a physical system in a covariant form [10], [20]. In contrast, there is still no consensus in GTR on how to calculate energy and momentum [9], [21-22].

In this regard, in [23], it was proposed to use the results of the covariant theory of gravitation in order to find the energy and momentum in GTR. To do this, it is desirable, as a first step, to replace the scalar fields used in GTR with the corresponding vector fields. Thus, in the Lagrangian density (2), the term  $-c^2\rho_0$  describes the scalar acceleration field, while the term  $-\rho_0\Pi$  refers to the scalar pressure field, taking into account the contribution of the elastic energy of the fluid. In contrast, each vector field is represented in the Lagrangian density by two terms rather than one. An example is the last two terms in (2), which describe the vector electromagnetic field. Thus, to pass to vector fields in (2), the term  $-c^2\rho_0$  should be replaced

by  $-U_\beta J^\beta - \frac{c^2}{16\pi\eta} u_{\alpha\beta} u^{\alpha\beta}$ , and the term  $-\rho_0\Pi$  should be replaced by  $-\pi_\beta J^\beta - \frac{c^2}{16\pi\sigma} f_{\alpha\beta} f^{\alpha\beta}$

, where  $U_\beta$  is the four-potential of acceleration field,  $J^\beta$  is the mass four-current,  $\eta$  is the coefficient of acceleration field,  $u_{\alpha\beta}$  is the tensor of acceleration field,  $\pi_\beta$  is the four-potential of pressure field,  $\sigma$  is the coefficient of pressure field,  $f_{\alpha\beta}$  is the tensor of pressure field [13].

Scalar pressure  $P$  is related to the specific energy  $\Pi$  of elastic compression of the fluid in relation (4) and contributes to the pressure four-force density in the equation of motion (23). In contrast, in the vector pressure field, the main role is played by the four-potential  $\pi_\beta$  and

the tensor  $f_{\alpha\beta}$ , while the pressure four-force density depends on the product  $J^\sigma f_{\sigma\mu}$ , where  $J^\sigma$  is the mass four-current [18-19].

Replacing the scalar acceleration field with a vector acceleration field, and replacing the scalar pressure field with a vector pressure field, will lead to significant changes in GTR. In this case, the stress-energy tensor  $\tau_{\mu\nu}$  (18) of the elastic fluid must be replaced with the sum of the stress-energy tensor  $B_{\mu\nu}$  of the vector acceleration field and the stress-energy tensor  $P_{\mu\nu}$  of the vector pressure field. As shown in [13], these tensors have a zero trace, so the following equalities hold:  $B_{\mu\nu}g^{\mu\nu} = 0$ ,  $P_{\mu\nu}g^{\mu\nu} = 0$ .

In contrast, in (20),  $\tau_{\mu\nu}g^{\mu\nu} \neq 0$ . This leads to the appearance of additional terms in the metric equation (22), which depend on the mass density and pressure, in addition to the sum of the tensors  $\tau_{\mu\nu} + W_{\mu\nu}$ . However, if the stress-energy tensors are correctly determined, they should be represented in the metric equation, and the additional terms should be absent. This follows from the very principle of constructing an equation for the metric, in which, on the one hand, there must be geometric quantities such as the Einstein tensor, and on the other hand, there must be stress-energy tensors as quantities that characterize the energy-dynamic properties of matter and fields. The presence of additional terms in (22) indicates that the Lagrangian density  $\mathcal{L}$  in (2) may be inaccurate and requires correction.

The advantage of using the vector acceleration field is that although the four-potential of this field is tied to the invariant mass density  $\rho_0$ , in GTR, the acceleration field can be given the additional property of existing outside of matter. In this case, an additional term  $\frac{c^2}{16\pi\eta}u_{\alpha\beta}u^{\alpha\beta}$  will appear in the integral in (61). By using this term in the general theory of relativity, we can determine the contribution of gravitational field outside the body to the system's energy, thereby solving the energy problem.

## 7. Conclusions

According to the approach adopted in the general theory of relativity, the equation of motion of the matter particles (23) follows from the relation  $\nabla^\nu T_{\mu\nu} = 0$  where the stress-energy tensor  $T_{\mu\nu}$  is presented in the equation for the metric (19). On the other hand, using the principle of least action, the equation of balance of the force densities acting on the matter particles is found in (30). Comparison of these two equations in Section 6 shows their complete



coincidence. However, the calculation of the energy outside a massive charged spherical body shows that only the electromagnetic field contributes to this energy. This clearly demonstrates the well-known problem of localization and calculation of gravitational energy in the general theory of relativity.

To solve this problem, it is proposed to replace the scalar acceleration field and the scalar pressure field in the general theory of relativity with corresponding vector fields, similar to what was done in the covariant theory of gravitation [10], [13], [15-20], [23].

### **Declarations**

#### **Availability of data and materials**

All data generated or analysed during this study are included in this published article.

#### **Competing interests**

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### **References**

1. Sharif M. and Fatima T. Energy-Momentum Distribution: A Crucial Problem in General Relativity. Int. J. Mod. Phys. A. Vol. 20 (18), 4309 (2005). <http://dx.doi.org/10.1142/S0217751X05020793>.
2. Crowther K. and De Haro S. Four Attitudes Towards Singularities in the Search for a Theory of Quantum Gravity. In Vassallo, Antonio (ed.). The Foundations of Spacetime Physics. New York (2022). <http://dx.doi.org/10.4324/9781003219019>.
3. Carroll S. The cosmological constant. Living Reviews in Relativity. Vol. 4 (1) 1 (2001). <http://dx.doi.org/10.12942/lrr-2001-1>.
4. Mendoza S. and Silva S. The matter Lagrangian of an ideal fluid. International Journal of Geometric Methods in Modern Physics, Vol. 18, No. 4, 2150059 (2021). <https://doi.org/10.1142/S0219887821500596>.
5. Diez-Tejedor A. Note on scalars, perfect fluids, constrained field theories, and all that. Physics Letters B, Vol. 727, pp. 27-30 (2013). <https://doi.org/10.1016/j.physletb.2013.10.030>.
6. Fock V.A. The Theory of Space, Time and Gravitation. Macmillan. (1964).

7. Minazzoli O. and Harko T. New derivation of the Lagrangian of a perfect fluid with a barotropic equation of state. *Phys. Rev. D*, Vol. 86, 087502 (2012).  
<https://doi.org/10.1103/PhysRevD.86.087502>.
8. Landau, L.D. and Lifshitz E.M. *The Classical Theory of Fields*. Vol. 2 (4th ed.). Butterworth-Heinemann, (1975). ISBN 978-0-7506-2768-9.
9. Szabados L.B. Quasi-Local Energy-Momentum and Angular Momentum in General Relativity. *Living Rev. Rel.* Vol. 12, 4 (2009) <https://doi.org/10.12942/lrr-2009-4>.
10. Fedosin S.G. What should we understand by the four-momentum of physical system? *Physica Scripta*, Vol. 99, No. 5, 055034 (2024). <https://doi.org/10.1088/1402-4896/ad3b45>.
11. Fedosin S.G. The generalized Poynting theorem for the general field and solution of the 4/3 problem. *International Frontier Science Letters*, Vol. 14, pp. 19-40 (2019).  
<https://zenodo.org/records/3252778>.
12. Landau L.D. and Lifshitz E.M. *Mechanics*. Vol. 1 (3rd ed.). Butterworth-Heinemann (1976). ISBN 978-0-7506-2896-9.
13. Fedosin S.G. About the cosmological constant, acceleration field, pressure field and energy. *Jordan Journal of Physics*, Vol. 9, No. 1, pp. 1-30 (2016).  
<http://dx.doi.org/10.5281/zenodo.889304>.
14. Nordström G. On the Energy of the Gravitational Field in Einstein's Theory. *Verhandl. Koninkl. Ned. Akad. Wetenschap., Afdel. Natuurk., Amsterdam*, Vol. 26, pp. 1201-1208 (1918).
15. Fedosin S.G. [The Integral Theorem of the Field Energy](#). *Gazi University Journal of Science*, Vol. 32, Issue 2, pp. 686-703 (2019). <https://doi.org/10.5281/zenodo.3252783>.
16. Fedosin S.G. Lagrangian formalism in the theory of relativistic vector fields. *International Journal of Modern Physics A*, Vol. 40, No. 02, 2450163 (2025).  
<https://doi.org/10.1142/S0217751X2450163X>.
17. Fedosin S.G. Two components of the macroscopic general field. *Reports in Advances of Physical Sciences*, Vol. 1, No. 2, 1750002 (2017).  
<http://dx.doi.org/10.1142/S2424942417500025>.
18. Fedosin S.G. Four-Dimensional Equation of Motion for Viscous Compressible and Charged Fluid with Regard to the Acceleration Field, Pressure Field and Dissipation Field. *International Journal of Thermodynamics*. Vol. 18, No. 1, pp. 13-24 (2015).  
<https://zenodo.org/records/3549371>.

19. Fedosin S.G. Equations of Motion in the Theory of Relativistic Vector Fields. International Letters of Chemistry, Physics and Astronomy, Vol. 83, pp. 12-30 (2019). <https://doi.org/10.18052/www.scipress.com/ILCPA.83.12>.
20. Fedosin S.G. Generalized Four-momentum for Continuously Distributed Materials. Gazi University Journal of Science, Vol. 37, Issue 3, pp. 1509-1538 (2024). <https://doi.org/10.35378/gujs.1231793>.
21. Sinya Aoki. Is energy conserved in general relativity? Modern Physics Letters A, Vol. 40, No. 40, 2550208 (2025). <https://doi.org/10.1142/S0217732325502086>.
22. Vieira W.M. and Magalhaes N.S. A strictly local solution to the problem of energy in general relativity. Frontiers in Astronomy and Space Sciences, Vol. 12, 1710235 (2026). <https://doi.org/10.3389/fspas.2025.1710235>.
23. Fedosin S.G. Comparison of tensor and vector theories of gravitation. Iranian Journal of Physics Research, Vol. 25, Issue 3, pp. 61-87 (2025). [https://ijpr.iut.ac.ir/article\\_3636\\_42baaa1a75b6c2906396b5b02f1bd012.pdf?lang=en](https://ijpr.iut.ac.ir/article_3636_42baaa1a75b6c2906396b5b02f1bd012.pdf?lang=en), <https://doi.org/10.47176/ijpr.25.1.01981>.

### Appendix A. Calculation of variation $\delta_x S$

The variation of the action function with respect to four-coordinate  $x^\mu$  in (27) has the following form:

$$\delta_x S = \frac{1}{c} \int \left( \begin{aligned} &c^2 \nabla_\beta (\rho_0 \delta x^\beta) - \rho_0 u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) + \\ &+ \frac{P}{\rho_0} \nabla_\beta (\rho_0 \delta x^\beta) - \frac{P}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) + \\ &+ \Pi \nabla_\beta (\rho_0 \delta x^\beta) - \frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) - \\ &- A_\beta \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\sigma} [\sqrt{-g} (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma)] \end{aligned} \right) \sqrt{-g} dx^0 dx^1 dx^2 dx^3. \quad (A1)$$

To transform the integrals in (A1), we need an expression for the four-divergence of an arbitrary four-vector  $B^\beta$  and an expression for the covariant derivative of an arbitrary four-vector  $A_\alpha$ :

$$\nabla_{\beta} B^{\beta} = \frac{1}{\sqrt{-g}} \frac{\partial (B^{\beta} \sqrt{-g})}{\partial x^{\beta}}. \quad (\text{A2})$$

$$\nabla_{\beta} A_{\alpha} = \frac{\partial A_{\alpha}}{\partial x^{\beta}} - \Gamma_{\alpha\beta}^{\lambda} A_{\lambda}. \quad (\text{A3})$$

For the first integral in (A1), using (A2), we find:

$$\begin{aligned} & \frac{1}{c} \int \left[ c^2 \nabla_{\beta} (\rho_0 \delta x^{\beta}) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\ & = c \int \frac{\partial (\rho_0 \delta x^{\beta} \sqrt{-g})}{\partial x^{\beta}} dx^0 dx^1 dx^2 dx^3 = c \int \rho_0 \delta x^{\beta} \sqrt{-g} d\Sigma_{\beta} = 0. \end{aligned} \quad (\text{A4})$$

In (A4), the integral over the four-dimensional volume of the four-divergence of the four-vector  $\rho_0 \delta x^{\beta}$  is transformed using the divergence theorem into an integral over the three-dimensional hypersurface. In (A4),  $d\Sigma_{\beta} = n_{\beta} d\Sigma$  is the orthonormal differential of the three-dimensional hypersurface  $\Sigma$  surrounding the physical system in the four-dimensional space;  $n_{\beta}$  is the four-dimensional normal vector perpendicular to the hypersurface  $\Sigma$  and pointing outward. The equality of the integral to zero in (A4) follows from the fact that on the hypersurface  $\Sigma$ , the variations  $\delta x^{\beta}$  are equal to zero according to the variation conditions.

The second integral in (A1) is transformed by parts:

$$\begin{aligned} & \frac{1}{c} \int \left[ -\rho_0 u_{\beta} u^{\sigma} \nabla_{\sigma} (\delta x^{\beta}) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\ & = -\frac{1}{c} \int \nabla_{\sigma} (\rho_0 u_{\beta} u^{\sigma} \delta x^{\beta}) \sqrt{-g} dx^0 dx^1 dx^2 dx^3 + \\ & + \frac{1}{c} \int \nabla_{\sigma} (\rho_0 u_{\beta} u^{\sigma}) \delta x^{\beta} \sqrt{-g} dx^0 dx^1 dx^2 dx^3. \end{aligned} \quad (\text{A5})$$

The first integral on the right-hand side of (A5) will not contribute to the variation result when integrating in (A1). This integral contains the four-divergence of the four-vector  $\rho_0 u_{\beta} u^{\sigma} \delta x^{\beta}$  and, given (A2) and the divergence theorem, is equal to zero, as the integral in (A4). Therefore, we can write:

$$\begin{aligned}
& \frac{1}{c} \int \left[ -\rho_0 u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int \nabla_\sigma (\rho_0 u_\beta u^\sigma) \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A6}$$

The third integral in (A1) transforms as follows:

$$\begin{aligned}
& \frac{1}{c} \int \left[ \frac{P}{\rho_0} \nabla_\beta (\rho_0 \delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int \nabla_\beta (P \delta x^\beta) \sqrt{-g} dx^0 dx^1 dx^2 dx^3 - \\
& - \frac{1}{c} \int \rho_0 \nabla_\beta \left( \frac{P}{\rho_0} \right) \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A7}$$

The first integral on the right-hand side of (A7), taking into account the divergence theorem, is equal to zero, similarly to (A4). Taking this into account, the following is obtained in (A7):

$$\begin{aligned}
& \frac{1}{c} \int \left[ \frac{P}{\rho_0} \nabla_\beta (\rho_0 \delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = -\frac{1}{c} \int \rho_0 \nabla_\beta \left( \frac{P}{\rho_0} \right) \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A8}$$

Let us transform the fourth integral in (A1):

$$\begin{aligned}
& \frac{1}{c} \int \left[ -\frac{P}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = -\frac{1}{c} \int \nabla_\sigma \left( \frac{P}{c^2} u_\beta u^\sigma \delta x^\beta \right) \sqrt{-g} dx^0 dx^1 dx^2 dx^3 + \\
& + \frac{1}{c} \int \nabla_\sigma \left( \frac{P}{c^2} u_\beta u^\sigma \right) \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A9}$$

The first integral on the right-hand side of (A9), taking into account (A2) and the divergence theorem, is equal to zero. This leads to the simplification of (A9):

$$\begin{aligned}
& \frac{1}{c} \int \left[ -\frac{P}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int \nabla_\sigma \left( \frac{P}{c^2} u_\beta u^\sigma \right) \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A10}$$

The transformation of the fifth integral in (A1) gives:

$$\begin{aligned}
& \frac{1}{c} \int \left[ \Pi \nabla_\beta (\rho_0 \delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int \nabla_\beta (\Pi \rho_0 \delta x^\beta) \sqrt{-g} dx^0 dx^1 dx^2 dx^3 - \frac{1}{c} \int \rho_0 \nabla_\beta \Pi \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A11}$$

The first integral on the right-hand side of (A11) is equal to zero, like (A4), since it contains the four-divergence of the four-vector  $\Pi \rho_0 \delta x^\beta$ . Taking this into account, we find in (A11):

$$\begin{aligned}
& \frac{1}{c} \int \left[ \Pi \nabla_\beta (\rho_0 \delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = -\frac{1}{c} \int \rho_0 \nabla_\beta \Pi \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A12}$$

The sixth integral in (A1) is transformed as follows:

$$\begin{aligned}
& \frac{1}{c} \int \left[ -\frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = -\frac{1}{c} \int \nabla_\sigma \left( \frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \delta x^\beta \right) \sqrt{-g} dx^0 dx^1 dx^2 dx^3 + \\
& + \frac{1}{c} \int \nabla_\sigma \left( \frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \right) \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A13}$$

The first integral on the right-hand side of (A13), taking into account the divergence theorem, is equal to zero, similar to (A4). The following remains in (A13):

$$\begin{aligned}
& \frac{1}{c} \int \left[ -\frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \nabla_\sigma (\delta x^\beta) \right] \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int \nabla_\sigma \left( \frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \right) \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A14}$$

The seventh integral in (A1) is transformed by parts as follows:

$$\begin{aligned}
& \frac{1}{c} \int \left\{ -A_\beta \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\sigma} \left[ \sqrt{-g} (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma) \right] \right\} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = -\frac{1}{c} \int \frac{\partial}{\partial x^\sigma} \left[ A_\beta \sqrt{-g} (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma) \right] dx^0 dx^1 dx^2 dx^3 + \\
& + \frac{1}{c} \int (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma) \frac{\partial A_\beta}{\partial x^\sigma} \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A15}$$

The first integral on the right-hand side of (A15), after applying the divergence theorem, becomes zero:

$$\begin{aligned}
& -\frac{1}{c} \int \frac{\partial}{\partial x^\sigma} \left[ A_\beta \sqrt{-g} (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma) \right] dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int \left[ A_\beta \sqrt{-g} (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma) \right] d\Sigma_\sigma = 0.
\end{aligned} \tag{A16}$$

In (A16)  $d\Sigma_\sigma = n_\sigma d\Sigma$  is the orthonormal differential of the three-dimensional hypersurface  $\Sigma$  surrounding the four-volume under consideration. The integral in (A16) is zero, since on the hypersurface  $\Sigma$  the variations  $\delta x^\beta$  and  $\delta x^\sigma$  are zero by the variation condition.

The second integral on the right-hand side of (A15), taking into account (A3), is transformed as follows:

$$\begin{aligned}
& \frac{1}{c} \int (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma) \frac{\partial A_\beta}{\partial x^\sigma} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int \left( \frac{\partial A_\beta}{\partial x^\sigma} - \frac{\partial A_\sigma}{\partial x^\beta} \right) j^\sigma \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\
& = \frac{1}{c} \int (\nabla_\sigma A_\beta - \nabla_\beta A_\sigma) j^\sigma \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \frac{1}{c} \int j^\sigma F_{\sigma\beta} \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3.
\end{aligned} \tag{A17}$$

In (A17), the definition of the electromagnetic field tensor is used in the form of  $F_{\sigma\beta} = \nabla_\sigma A_\beta - \nabla_\beta A_\sigma$ .

From (A15-A17), we find:

$$\begin{aligned} & \frac{1}{c} \int \left\{ -A_\beta \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\sigma} \left[ \sqrt{-g} (j^\sigma \delta x^\beta - j^\beta \delta x^\sigma) \right] \right\} \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = \\ & = \frac{1}{c} \int j^\sigma F_{\sigma\beta} \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3. \end{aligned} \quad (\text{A18})$$

We substitute the expressions (A4), (A6), (A8), (A10), (A12), (A14), (A18) into (A1) and set the variation  $\delta_x S$  equal to zero:

$$\delta_x S = \frac{1}{c} \int \left[ \begin{aligned} & \nabla_\sigma (\rho_0 u_\beta u^\sigma) - \rho_0 \nabla_\beta \left( \frac{P}{\rho_0} \right) + \nabla_\sigma \left( \frac{P}{c^2} u_\beta u^\sigma \right) - \\ & - \rho_0 \nabla_\beta \Pi + \nabla_\sigma \left( \frac{\rho_0 \Pi}{c^2} u_\beta u^\sigma \right) + j^\sigma F_{\sigma\beta} \end{aligned} \right] \delta x^\beta \sqrt{-g} dx^0 dx^1 dx^2 dx^3 = 0. \quad (\text{A19})$$

For the variation  $\delta S_2$  to be zero in (A19), the square bracket in (A19) must be zero. This gives the equation:

$$\nabla_\sigma \left[ \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u_\beta u^\sigma \right] - \rho_0 \nabla_\beta \left( \frac{P}{\rho_0} \right) - \rho_0 \nabla_\beta \Pi + j^\sigma F_{\sigma\beta} = 0. \quad (\text{A20})$$

After rewriting the second term in (A20) as

$$-\rho_0 \nabla_\beta \left( \frac{P}{\rho_0} \right) = -\nabla_\beta P + \frac{P}{\rho_0} \nabla_\beta \rho_0. \quad (\text{A21})$$

and replacing the index  $\beta$  with the index  $\mu$  the equation (A20) can be written as follows:



$$\nabla_\sigma \left[ \left( \rho_0 + \frac{\rho_0 \Pi + P}{c^2} \right) u_\mu u^\sigma \right] - \nabla_\mu P + \frac{P}{\rho_0} \nabla_\mu \rho_0 - \rho_0 \nabla_\mu \Pi + j^\sigma F_{\sigma\mu} = 0. \quad (\text{A22})$$

### Appendix B. Calculation of scalar curvature outside the body

According to (37), the equation for the metric outside a stationary charged spherical body is as follows:

$$R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu} = \frac{8\pi G\beta}{c^4} W_{\mu\nu}. \quad (\text{B1})$$

Next, we assume that the mass density  $\rho_0$  and charge density  $\rho_{0q}$  are constant throughout the body, there are no mass or electric currents, and the intrinsic motion of the matter particles can be neglected. In this case, the vector potential and magnetic field of the body are zero, and only the electric field strength  $\mathbf{E}$  contributes to the stress-energy tensor  $W_{\mu\nu}$ . In spherical coordinates  $q^\mu = ct, r, \theta, \phi$ , the four-potential  $A_\mu$  and electromagnetic field tensor  $F_{\mu\nu}$  can be written as follows:

$$A_\mu = \left( \frac{\varphi}{c}, -\mathbf{A} \right) = \left( \frac{\varphi}{c}, 0, 0, 0 \right) = \left( \frac{Q}{4\pi\epsilon_0 cr}, 0, 0, 0 \right).$$

$$F_{\mu\nu} = \frac{\partial A_\nu}{\partial q^\mu} - \frac{\partial A_\mu}{\partial q^\nu} = \begin{pmatrix} 0 & \frac{Q}{4\pi\epsilon_0 cr^2} & 0 & 0 \\ -\frac{Q}{4\pi\epsilon_0 cr^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{B2})$$

The metric tensor outside a stationary spherical massive body with a centrally symmetric mass distribution is as follows [8]:

$$g_{\mu\nu} = \begin{pmatrix} B & 0 & 0 & 0 \\ 0 & -K & 0 & 0 \\ 0 & 0 & -E & 0 \\ 0 & 0 & 0 & -E \sin^2 \theta \end{pmatrix}, \quad (\text{B3})$$

$$g^{\lambda\eta} = \begin{pmatrix} \frac{1}{B} & 0 & 0 & 0 \\ 0 & -\frac{1}{K} & 0 & 0 \\ 0 & 0 & -\frac{1}{E} & 0 \\ 0 & 0 & 0 & -\frac{1}{E \sin^2 \theta} \end{pmatrix}, \quad (\text{B4})$$

Using (B2) and (B4), we can calculate  $F^{\lambda\eta}$ :

$$F^{\lambda\eta} = F_{\mu\nu} g^{\mu\lambda} g^{\nu\eta} = \begin{pmatrix} 0 & -\frac{Q}{4\pi\epsilon_0 cr^2 BK} & 0 & 0 \\ \frac{Q}{4\pi\epsilon_0 cr^2 BK} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{B5})$$

Substituting (B2-B5) into the expression for the tensor  $W_{\mu\nu}$ , and taking into account the relation  $\mu_0 \epsilon_0 c^2 = 1$ , we find:

$$W_{\mu\nu} = \frac{1}{\mu_0} \left( -F_{\alpha\nu} F_{\lambda\mu} g^{\lambda\alpha} + \frac{1}{4} F_{\alpha\beta} F^{\alpha\beta} g_{\mu\nu} \right) = \begin{pmatrix} \frac{Q^2}{32\pi^2 \epsilon_0 r^4 K} & 0 & 0 & 0 \\ 0 & -\frac{Q^2}{32\pi^2 \epsilon_0 r^4 B} & 0 & 0 \\ 0 & 0 & \frac{Q^2 E}{32\pi^2 \epsilon_0 r^4 BK} & 0 \\ 0 & 0 & 0 & \frac{Q^2 E \sin^2 \theta}{32\pi^2 \epsilon_0 r^4 BK} \end{pmatrix}. \quad (\text{B6})$$

To calculate the Christoffel symbols, the following expression is used:

$$\Gamma_{\beta\mu}^{\sigma} = \frac{1}{2} g^{\sigma\nu} \left( \frac{\partial g_{\mu\nu}}{\partial q^{\beta}} + \frac{\partial g_{\nu\beta}}{\partial q^{\mu}} - \frac{\partial g_{\beta\mu}}{\partial q^{\nu}} \right). \quad (\text{B7})$$

In spherical coordinates  $q^{\mu} = ct, r, \theta, \phi$ , the non-zero Christoffel symbols are expressed in (B7) through the functions  $B$ ,  $K$ ,  $E$ , contained in the components of the metric tensor (B3-B4):

$$\begin{aligned} \Gamma_{01}^0 = \Gamma_{10}^0 = \frac{B'}{2B}, \quad \Gamma_{00}^1 = \frac{B'}{2K}, \quad \Gamma_{11}^1 = \frac{K'}{2K}, \quad \Gamma_{22}^1 = -\frac{E'}{2K}, \quad \Gamma_{33}^1 = -\frac{E' \sin^2 \theta}{2K}, \\ \Gamma_{12}^2 = \Gamma_{21}^2 = \Gamma_{13}^3 = \Gamma_{31}^3 = \frac{E'}{2E}, \quad \Gamma_{33}^2 = -\sin \theta \cos \theta, \quad \Gamma_{23}^3 = \Gamma_{32}^3 = \text{ctg} \theta. \end{aligned} \quad (\text{B8})$$

When calculating in (B8), it was taken into account that the functions  $B$ ,  $K$ ,  $E$  depend only on the radial component  $r$ , and the components of the metric tensor (B3-B4) do not depend on the time  $t$  and on the angle  $\phi$ . For the derivatives in (B8), the following notations are used:  $B' = \frac{dB}{dr}$ ,  $K' = \frac{dK}{dr}$ ,  $E' = \frac{dE}{dr}$ .

The Ricci tensor  $R_{\mu\alpha}$  is calculated either through the metric tensor  $g^{\nu\beta}$  and the curvature tensor  $R_{\nu\mu\beta\alpha}$ , or using the Christoffel symbols:

$$R_{\mu\alpha} = g^{\nu\beta} R_{\nu\mu\beta\alpha} = \frac{\partial \Gamma_{\mu\alpha}^{\beta}}{\partial q^{\beta}} - \frac{\partial \Gamma_{\mu\beta}^{\alpha}}{\partial q^{\alpha}} + \Gamma_{\mu\alpha}^{\lambda} \Gamma_{\lambda\beta}^{\beta} - \Gamma_{\mu\beta}^{\lambda} \Gamma_{\lambda\alpha}^{\beta}. \quad (\text{B9})$$

Using the Christoffel symbols (B8) in (B9), we find the components of the Ricci tensor  $R_{\mu\alpha}$ :

$$R_{00} = \frac{1}{2} \frac{\partial}{\partial r} \left( \frac{B'}{K} \right) + \frac{B'}{4K} \left( \frac{K'}{K} + \frac{2E'}{E} - \frac{B'}{B} \right).$$

$$R_{11} = -\frac{1}{2} \frac{\partial}{\partial r} \left( \frac{B'}{B} \right) - \frac{\partial}{\partial r} \left( \frac{E'}{E} \right) + \frac{K'}{4K} \left( \frac{B'}{B} + \frac{2E'}{E} \right) - \frac{B'^2}{4B^2} - \frac{E'^2}{2E^2}.$$

$$R_{22} = -\frac{1}{2} \frac{\partial}{\partial r} \left( \frac{E'}{K} \right) - \frac{E'}{4K} \left( \frac{B'}{B} + \frac{K'}{K} \right) + 1.$$

$$R_{33} = -\frac{1}{2} \frac{\partial}{\partial r} \left( \frac{E' \sin^2 \theta}{K} \right) - \frac{E' \sin^2 \theta}{4K} \left( \frac{B'}{B} + \frac{K'}{K} \right) + \sin^2 \theta.$$

(B10)

Using the components  $g^{\lambda\eta}$  (B4) and the components  $R_{\lambda\eta}$  (B10), we find the scalar curvature  $R$ :

$$R = g^{\lambda\eta} R_{\lambda\eta} = \frac{1}{2B} \frac{\partial}{\partial r} \left( \frac{B'}{K} \right) + \frac{1}{2K} \frac{\partial}{\partial r} \left( \frac{B'}{B} \right) + \frac{1}{K} \frac{\partial}{\partial r} \left( \frac{E'}{E} \right) + \frac{1}{E} \frac{\partial}{\partial r} \left( \frac{E'}{K} \right) + \frac{B'E'}{BKE} + \frac{E'^2}{2KE^2} - \frac{2}{E}. \quad (B11)$$

Substituting the components of the Ricci tensor  $R_{\mu\nu}$  (B10), the scalar curvature  $R$  (B11), the components of the metric tensor  $g_{\mu\nu}$  (B3), and the components of the stress-energy tensor  $W_{\mu\nu}$  (B6) into the metric equation (B1), we obtain four equations:

$$\frac{B''}{4} - \frac{BE''}{2E} - \frac{B'^2}{8B} - \frac{B'K'}{8K} + \frac{B'E'}{4E} + \frac{BE'^2}{8E^2} + \frac{BK'E'}{4KE} + \frac{BK}{2E} = \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^4}. \quad (B12)$$

$$-\frac{B''}{4} - \frac{BE''}{2E} + \frac{B'^2}{8B} + \frac{B'K'}{8K} + \frac{B'E'}{4E} + \frac{3BE'^2}{8E^2} + \frac{BK'E'}{4KE} - \frac{BK}{2E} = -\frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^4}. \quad (B13)$$

$$\frac{B''}{4} - \frac{B'K'}{8K} - \frac{B'^2}{8B} - \frac{BE'^2}{8E^2} + \frac{BK}{2E} = \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^4}. \quad (B14)$$

$$\frac{B''}{4} - \frac{B'K'}{8K} - \frac{B'^2}{8B} - \frac{BE'^2}{8E^2} + \frac{BK}{2E} = \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^4}. \quad (\text{B15})$$

In (B12-B15), the following notations are made:  $B'' = \frac{dB'}{dr} = \frac{d^2 B}{dr^2}$ ,  $E'' = \frac{dE'}{dr} = \frac{d^2 E}{dr^2}$ .

By adding equations (B12) and (B13) and transforming the result, we find:

$$-\frac{BE''}{E} + \frac{B'E'}{2E} + \frac{BE'^2}{2E^2} + \frac{BK'E'}{2KE} = 0. \quad \frac{B'}{B} + \frac{E'}{E} + \frac{K'}{K} = \frac{2E''}{E'}. \quad (\text{B16})$$

$$d \ln(BKE) = d \ln E'^2. \quad \frac{E'^2}{BKE} = C_1.$$

In (B16),  $C_1$  is a constant. At infinity, far from the massive body under consideration, the components of the metric tensor  $g_{\mu\nu}$  (B3) must equal the components of the metric tensor  $\eta_{\mu\nu}$  of Minkowski's flat spacetime. In this case, in spherical coordinates, we have  $B=1$ ,  $K=1$ ,  $E=r^2$ , and then in (B16), we see that  $C_1 = 4$ .

From (B16), we can express the function  $K$  and  $\frac{K'}{K}$ , given that  $C_1 = 4$ :

$$K = \frac{E'^2}{4BE}. \quad \frac{K'}{K} = \frac{2E''}{E'} - \frac{B'}{B} - \frac{E'}{E}. \quad (\text{B17})$$

Substituting (B17) into (B12-B15) gives the same equation:

$$B'' - \frac{B'E''}{E'} + \frac{B'E'}{2E} - \frac{BE'^2}{2E^2} + \frac{E'^2}{2E^2} = \frac{Q^2 G \beta}{\pi \epsilon_0 c^4 r^4}. \quad (\text{B18})$$

Comparing equations (B14) and (B15) shows that these equations are equal to each other. Additionally, equation (B12) is equal to the sum of equations (B13), (B14), and (B15). This means that among the equations (B12), (B13), and (B14), only two equations are independent,

and these equations are necessary to determine the three unknown functions  $B, K, E$ . In this regard, we simplify the problem and assume that  $E = r^2$ .

Since the interval in spherical coordinates, taking into account the expression  $g_{\mu\nu}$  (B3), is written as

$$ds = \sqrt{g_{\mu\nu} dq^\mu dq^\nu} = \sqrt{c^2 B dt^2 - K dr^2 - E d\theta^2 - E \sin^2 \theta d\phi^2}, \quad (\text{B19})$$

then, according to (B19) with  $E = r^2$ , displacements in directions perpendicular to the radial coordinate  $r$ , contribute to the interval in the same way as in the flat Minkowski spacetime. This means that when the condition  $E = r^2$  is selected, the dimensions of bodies that are perpendicular to the radial coordinate  $r$  remain unchanged in the curved spacetime. This applies, in particular, to the circumference on the surface of the sphere and to the formula for calculating it.

For  $E = r^2$  in (B17), we find:

$$K = \frac{1}{B}, \quad \frac{K'}{K} = -\frac{B'}{B}. \quad (\text{B20})$$

Substituting  $E = r^2$  in (B18) gives an equation for  $B$ :

$$B'' - \frac{2B}{r^2} + \frac{2}{r^2} = \frac{Q^2 G \beta}{\pi \varepsilon_0 c^4 r^4}. \quad (\text{B21})$$

The solution of equation (B21) with zero right-hand side is

$$B_1 = 1 + \frac{C_2}{r} + C_3 r^2, \quad (\text{B22})$$

where  $C_2$  and  $C_3$  are some constants.

A particular solution of the equation (B21) with a nonzero right-hand side is

$$B_2 = 1 + \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^2}. \quad (\text{B23})$$

In (B22), the constant  $C_3$  must be equal to zero, otherwise the function  $B_1$  will be infinite for large values of  $r$ . Taking this into account, the function  $B$  is obtained as a combination of the solutions  $B_1$  (B22) and  $B_2$  (B23):

$$B = 1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^2}. \quad (\text{B24})$$

From (B20) and (B24), it follows

$$K = \frac{1}{B} = \frac{1}{1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^2}}. \quad (\text{B25})$$

Given (B24-B25) and  $E = r^2$ , we find the components of the metric tensor (B3):

$$\begin{aligned} g_{00} &= B = 1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^2}, & g_{11} &= -K = -\frac{1}{1 + \frac{C_2}{r} + \frac{Q^2 G \beta}{4\pi \epsilon_0 c^4 r^2}}. \\ g_{22} &= -E = -r^2, & g_{33} &= -E \sin^2 \theta = -r^2 \sin^2 \theta. \end{aligned} \quad (\text{B26})$$

Substituting  $B, K, E$  from (B26) into (B11) allows us to find the scalar curvature  $R$  outside the body:

$$R = 0. \quad (\text{B27})$$